

5(a)	plasmodesma ; <i>plus any one from</i> R if stated as apoplast 1 allows / AW, <u>symplast</u>, pathway / movement ; 2 allows transport of / AW, water / sucrose / sugars / amino acids / organic substances / ions / mineral salts (A minerals) / lipids / assimilates / hormones / cell signalling molecules ; <i>in context of</i> between cells I substance unqualified I osmosis of water if in a longer list where other substances are named correctly R osmosis of water <i>if this is the only item noted</i> 3 cell to cell / AW, communication / signalling ; A cell communication	2
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Question	Answer	Marks
5(b)(i)	<i>any one from</i> (only) polypeptides / proteins, synthesised at ribosomes ; carbohydrates / polysaccharides, not synthesised at ribosomes ; enzymes, for synthesis / required, not located at ribosomes ;	1
5(b)(ii)	exocytosis ;	1

Question	Answer	Marks
5(c)	<i>accept substrate for cellulose and enzyme for cellulase</i> <i>allow use of extracted data to support a response to confirm awarding a mark</i> 1 correct description ; <i>overall trend or more detailed break down</i> e.g. as, cellulose / substrate, concentration increases, rate of reaction increases or as, cellulose / substrate, concentration increases, steep increase in rate of reaction then, less steep increase / (begins to) plateau I 'rate of reaction is fast and then slows down' <i>needs ref. to rate of reaction, describing 'gradient of curve' is not enough</i> <i>plus any two from</i> 2 increasing cellulose concentration increases number of, collisions between enzyme and substrate / enzyme-substrate complexes / ESCs ; A collisions between active site and substrate A idea of increased chances of cellulose binding to active site 3 at low cellulose concentrations, (many) active sites, available / not all used / not saturated ; 4 at higher cellulose concentrations, active sites becoming saturated / most active sites occupied / most active sites not available / AW ; A active sites are, saturated / occupied 5 named limiting factor(s), qualified ; e.g. at low(er) cellulose concentrations, cellulose concentration limiting / enzyme concentration not limiting A idea of, not enough cellulose / enzyme in excess as cellulose concentration increases, change from cellulose concentration limiting to enzyme concentration as limiting factor at higher cellulose concentrations, enzyme concentration limiting / cellulose concentration not limiting A idea of, cellulose in excess / not enough enzyme / enzyme working at its fastest rate / active sites not emptying quickly enough for substrate available	3

Question	Answer	Marks
5(d)(i)	<i>mark to max 2 if microorganisms are considered to be enzyme inhibitors</i> <i>lower K_m for pectinase, any two from ultrasound</i> increases affinity of enzyme for substrate ; makes shape of active site more complementary ; makes (position of) active site more accessible (to substrate) ; A makes it easier for substrate to enter (active site) breaks up pectin, aggregates / AW, to expose more substrate for binding ; <i>higher V_{max} for xylanase, any two from ultrasound</i> increases rate of, collision between enzyme and substrate / more enzyme-substrate complexes per unit time ; I more ESC form <i>needs idea of increased rate</i> increases rate of catalysis after binding ; may change substrate for easier hydrolysis ; may lower, more than normal, activation energy required for reaction ; A described, e.g. increased ability to, break / form, bonds	3
5(d)(ii)	yes (as all have) increased, catalytic efficiency / productivity / V_{max}/K_m ; A manipulated data for increased catalytic efficiency any <u>two</u> from pectin 24.5 / 25% cellulase 17.25 / 17% xylanase 16.8 / 17%	1